

Healthcare Service Utilization Analysis Using Python

1.SUMMARY

This project analyzes Healthcare Service Utilization data from **California** between **2021** to **2023** using **Python**. The analysis focuses on identifying **healthcare utilization** patterns across service categories, payer types, demographic groups, and regions. Data preprocessing, statistical analysis, feature engineering, and data visualization were performed to generate **meaningful insights** that support healthcare planning, resource allocation, and data-driven decision-making.

2.BUSINESS PROBLEM

Healthcare organizations generate **large volumes** of service utilization data, making it **challenging** to identify **patterns** in healthcare usage across different service categories, payer types, demographic groups, and regions. Without **proper analysis**, healthcare providers and policymakers may face **difficulties** in optimizing resource allocation and planning effective healthcare services. This project addresses these challenges by **analyzing** healthcare service utilization data from **2021** to **2023**, providing insights that support informed decision-making and improve healthcare planning.

3.PROJECT OBJECTIVES

- To identify the most utilized healthcare service categories across **California** between **2021** to **2023**.
- To compare healthcare service utilization across different **payer types** to understand variations in service demand.
- To analyze demographic and regional variations in healthcare service utilization based on **age group**, **assigned sex at birth**, and **Covered California regions**.
- To evaluate insurance coverage patterns by analyzing **medical coverage months** across **different payer types** and **counties**.
- To identify healthcare utilization trends between **2021** to **2023** and examine changes in service demand across different healthcare service categories.
- To generate actionable insights that support healthcare **planning**, **resource allocation**, and **data-driven decision-making**.

4.DATASET INFORMATION

SOURCE: California Health and Human Services (CalHHS) – Healthcare Payments Data (HPD) Services Report

LOCATION: California, USA

YEAR/TIMELINE: 2021–2023 (Filtered from the original dataset covering 2018–2023)

DOMAIN: Healthcare Analytics

5.TOOLS AND TECHNOLOGIES

TOOL/TECHNOLOGY	PURPOSE
Python	Programming language used for data cleaning, preprocessing, feature engineering, statistical analysis, and data visualization.
Google Colab	Cloud-based platform used to develop, execute, and document the project.
Pandas	Used for data manipulation, cleaning, filtering, preprocessing, and analysis.
Numpy	Used for numerical computations and mathematical operations.
Matplotlib	Used to create charts and visualizations for exploratory data analysis.
Seaborn	Used to create statistical visualizations, including histograms for analyzing data distribution.

6.Project Methodology and Implementation

- Data Collection
- Initial Data Inspection
- Data Cleaning
- Feature Engineering
- Statistical Analysis
- Data Visualization (Exploratory Data Analysis)
- Business Insights & Findings

7.ATTRIBUTE DETAILS

ATTRIBUTE NAME	DESCRIPTION	DATA TPE
reporting_year	Year in which the healthcare services were reported.	Integer
payer_type	Type of health insurance or payer responsible for healthcare coverage (e.g., Commercial, Medicare, Medi-Cal).	Object
service_category	Category of healthcare service provided (e.g., E&M, Test, Imaging).	Object

member_count	Total number of members enrolled under a specific payer and service category.	Float
medical_coverage_months	Total number of months members were covered under a healthcare plan.	Float
service_count	Total number of healthcare services utilized.	Float
county_name	Name of the California county where the healthcare services were reported.	Object
covered_california_region	Covered California regional classification of each county.	Object
age_band	Age group of the population receiving healthcare services.	Object
assigned_sex_at_birth	Sex assigned at birth of the members (Male or Female).	Object
average_services_per_member	Feature created by dividing service_count by member_count.	Float
service_utilization_rate	Feature created by dividing service_count by medical_coverage_months.	Float

8.DATA PREPROCESSING

8.1 DATA CLEANING

- Filtered the dataset to include records from **2021** to **2023**, reducing the dataset from the original data to **120,960** rows and **12** columns since we are analyzing the **recent year** data.
- Verified the shape of the filtered dataset, which contained **60,480** rows and **12** columns before further data cleaning and preprocessing steps.
- Removed rows containing **zero** values where required, resulting in a final cleaned dataset of **57,555** rows and **12** columns.
- Removed rows containing **zero** values where required to improve **data quality**.
- Removed the **non-essential** columns **county_name_plus_spa** and **suppression_ind**.
- Checked the dataset **structure**, **shape**, **missing values**, and **duplicate** records at each stage of preprocessing.
- After removing the non-essential columns, **262** rows appeared as **duplicates**. These were not **actual** duplicate records in the original dataset but resulted from removing the **distinguishing** columns. Therefore, these records were retained for **analysis**.

- Analyzed the distribution of the **numerical** columns before selecting an **imputation** technique.
- Handled missing values in **member_count**, **medical_coverage_months**, and **service_count** using the group-wise **median** based on **payer_type** and **service_category**.
- Verified that all **missing values** were successfully handled after imputation.
- Checked the dataset again to ensure its **shape**, **duplicate count**, and **missing values** were correct.
- Reviewed the **data types** and confirmed that all attributes were already in the **correct format**, therefore no data type conversion was required.

8.2 FEATURE ENGINEERING

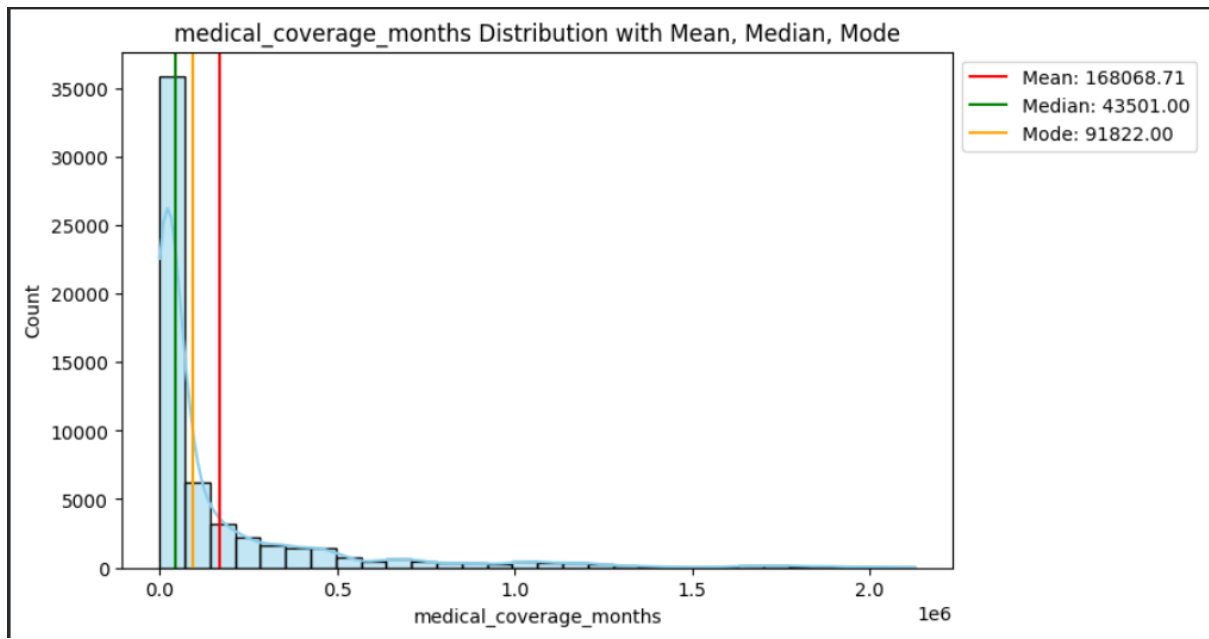
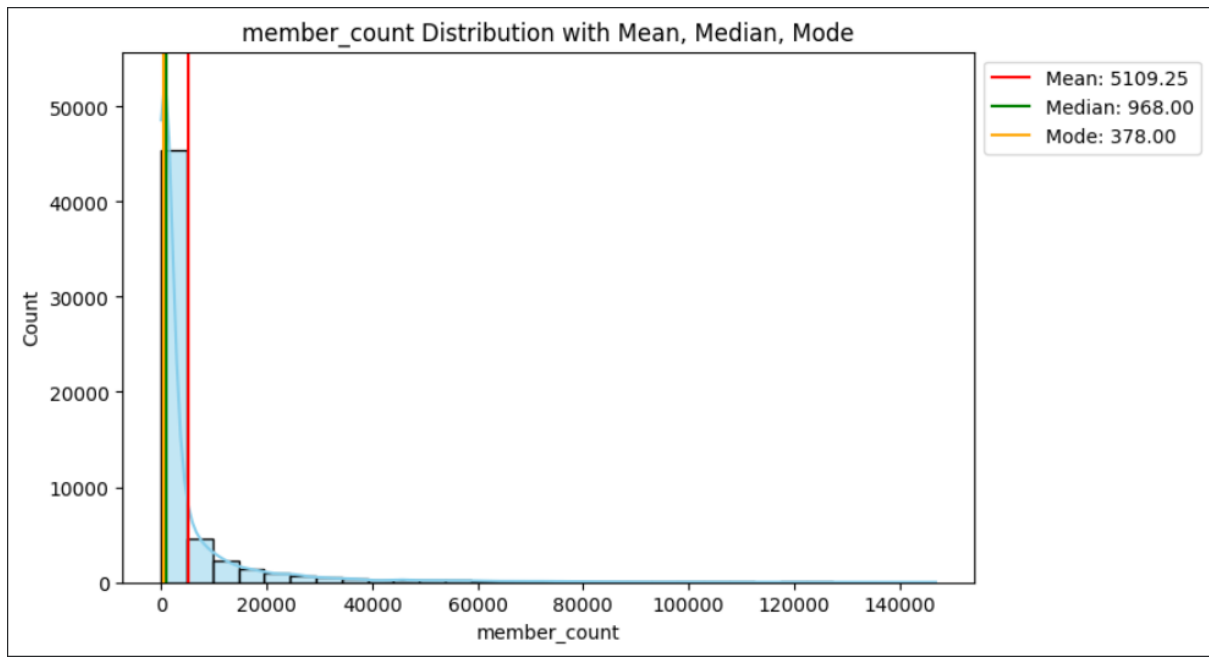
- Created the **avg_services_per_member** feature by **dividing service_count** by **member_count** to calculate the average number of healthcare services utilized per member.
- Verified that creating the **avg_services_per_member** feature did not introduce any **additional duplicate records** and confirmed the generated values.
- Created the **utilization_level** feature by **categorizing avg_services_per_member** into **Low**, **Medium**, and **High** utilization levels using quantile-based (**qcut**) binning.
- Verified that creating the **utilization_level** feature did not introduce any **additional** duplicate records and confirmed that the generated categories were **Low**, **Medium**, and **High**.
- Used the engineered features to support **deeper analysis** and improve the interpretation of healthcare service utilization patterns.

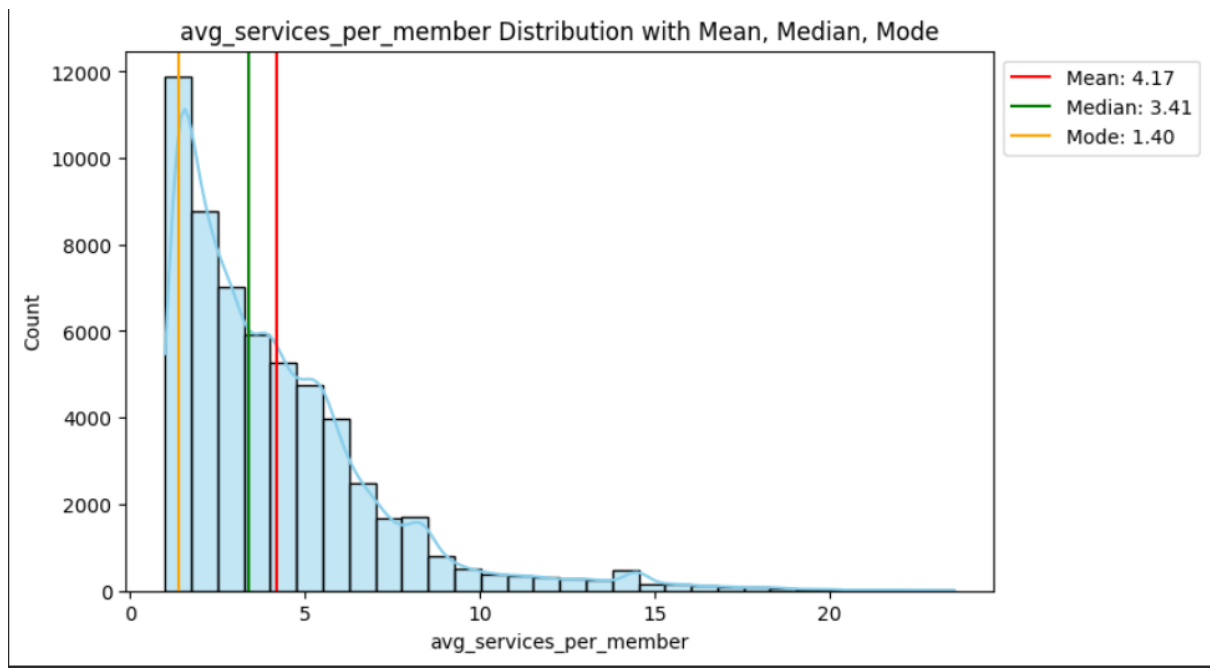
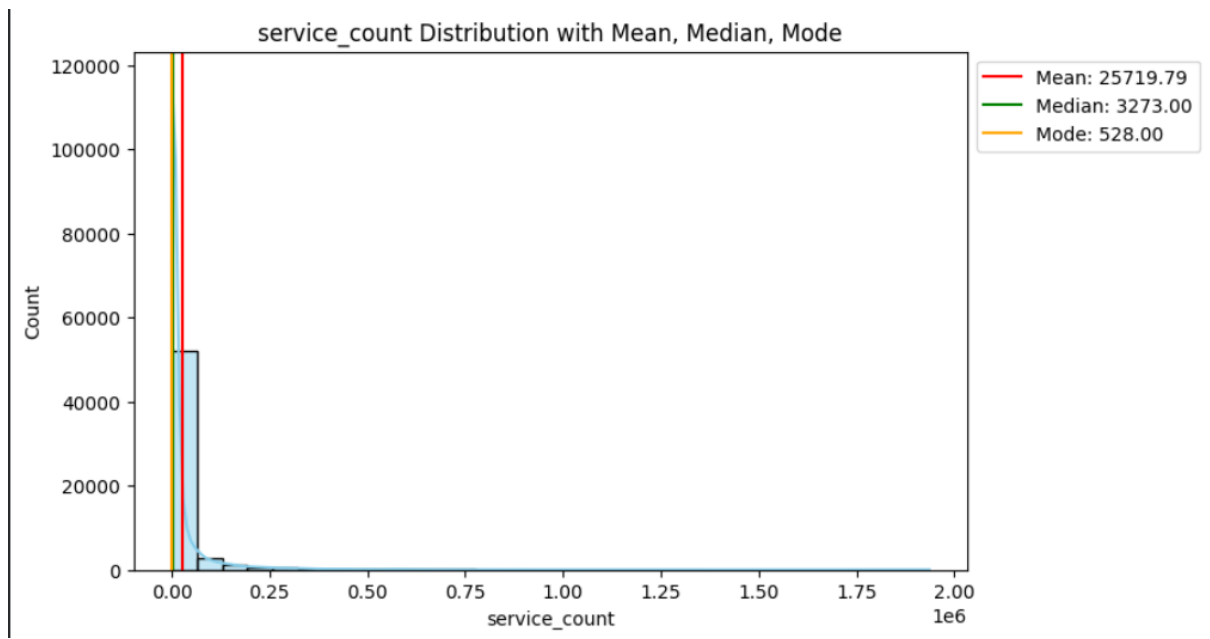
9. STATISTICAL ANALYSIS

9.1 MEASURE OF CENTRAL TENDENCY

Variable	Mean	Median	Mode
member_count	5,109.25	968.00	378.00
medical_coverage_months	168,068.71	43,501.00	91,822.00
service_count	25,719.79	3,273.00	528.00
avg_services_per_member	4.17	3.41	1.40

CENTRAL TENDENCY & DISTRIBUTION ANALYSIS





INTERPRETATION

- **The Blue Bars (Histogram):** Show the distribution of healthcare service utilization metrics across different groups.
- **The Blue Curve (KDE):** Displays the overall distribution pattern and confirms that most observations are concentrated at lower values.
- **The Orange Line (Mode):** Represents the most frequently occurring value in each healthcare metric.
- **The Green Line (Median):** Represents the middle value of the dataset and provides a reliable measure of the typical healthcare utilization.

- **The Red Line (Mean):** Represents the mathematical average. In all four variables, the Mean lies to the right of the Median, indicating positively skewed distributions. This suggests that a small number of records with exceptionally high values are increasing the overall averages.

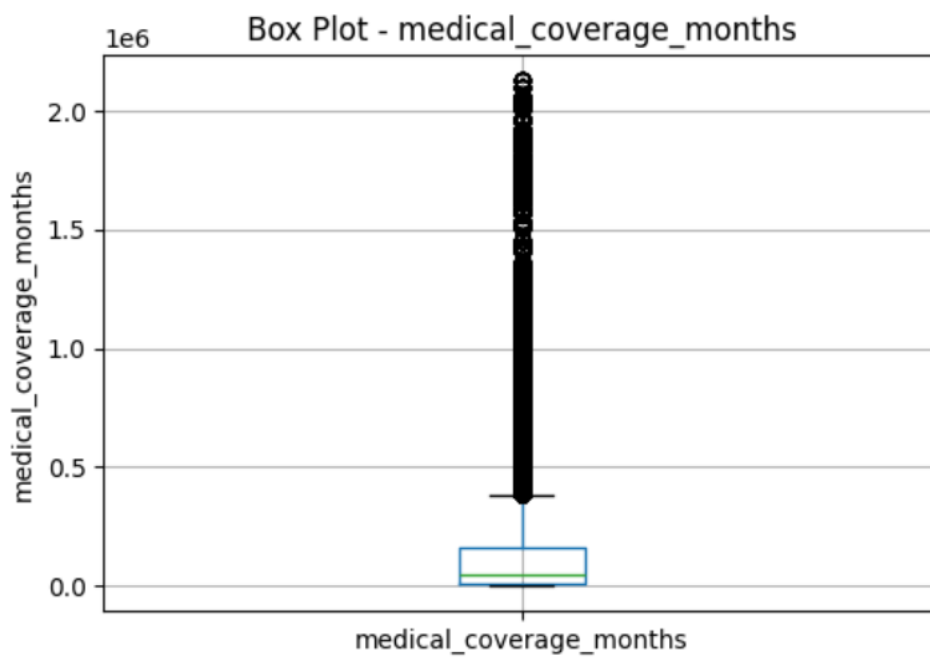
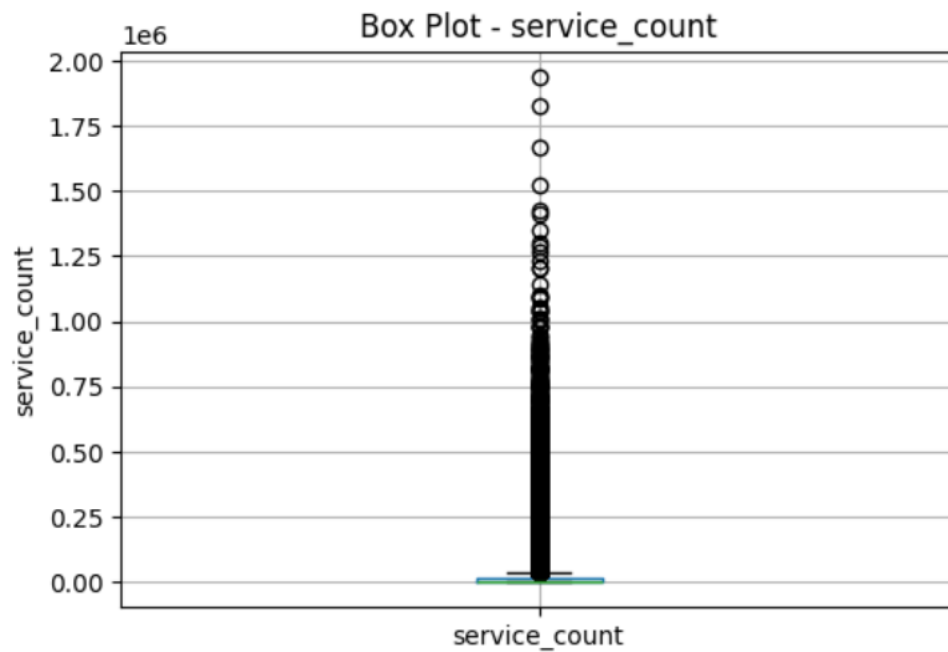
INSIGHTS

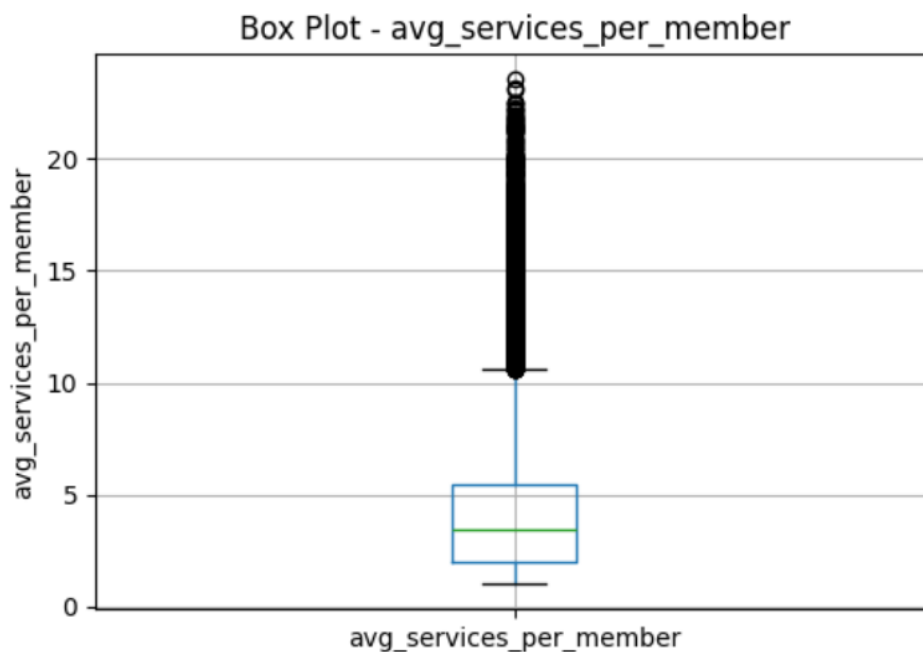
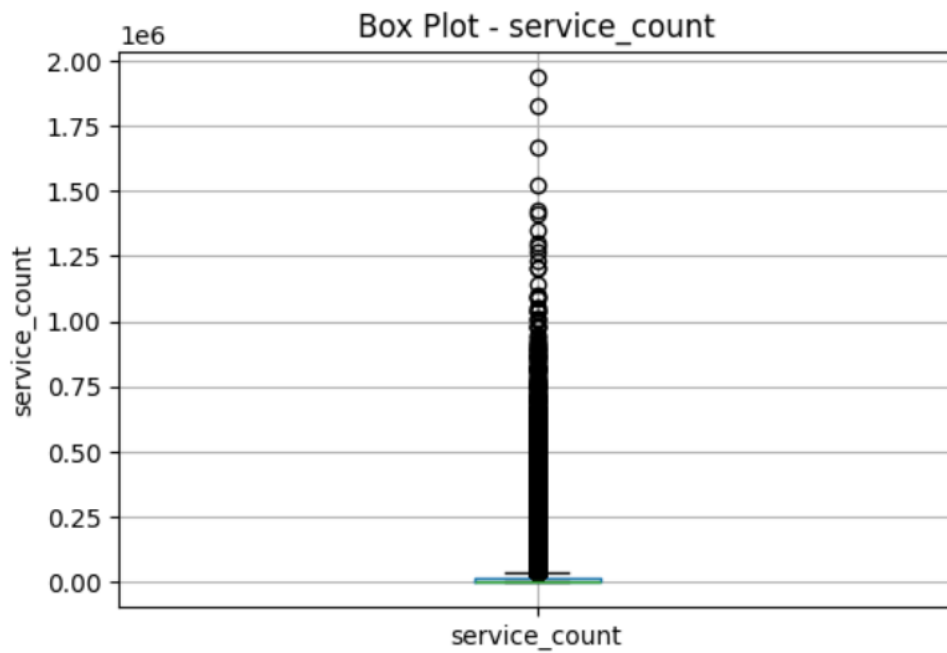
- **Member Count:** Most healthcare groups contain relatively fewer members, while a few groups have very large populations. The mean **(5,109.25)** is higher than the median **(968.00)** and mode **(378.00)**, indicating a right-skewed distribution.
- **Medical Coverage Months:** Most observations have lower coverage months, while a few groups record exceptionally high values. The mean **(168,068.71)** exceeds the median **(43,501.00)** and mode **(91,822.00)**, indicating positive skewness.
- **Service Count:** Most groups utilize relatively fewer healthcare services, while a limited number record very high service usage. The mean **(25,719.79)** is much higher than the median **(3,273.00)** and mode **(528.00)**.
- **Average Services per Member:** Most members utilize healthcare services at a moderate rate, while a few have substantially higher utilization. The mean **(4.17)** is greater than the median **(3.41)** and mode **(1.40)**, indicating positive skewness.
- **Overall Distribution:** All four variables exhibit **positive** (right) skewness, where the mean is greater than the median, indicating that high-value observations increase the overall average.
- **Possible Outliers:** The long right tails in the histograms suggest potential **outliers**, which should be verified using a Box Plot or the IQR method before any treatment.

9.3 MEASURE OF DISPERSION: VARIANCE

VARIABLE	VARIANCE
member_count	147,072,363.64
medical_coverage_months	94,925,221,933.13
service_count	5,679,367,784.22
avg_services_per_member	8.90

BOX PLOT





INTERPRETATION

- The **Variance** measures how much each healthcare utilization metric varies from its **average** value. **Higher** variance indicates greater variability among healthcare groups, while **lower** variance indicates more consistent values.
- The **Box Plot** visualizes the **distribution** of each variable and helps identify the **spread** of the data, median, and **potential outliers**. The presence of observations beyond the whiskers suggests potential outliers in the dataset.

INSIGHTS

- **Member Count:** The box plot shows several upper outliers, indicating large differences in member populations. The variance **(147,072,363.64)** confirms a wide spread in member counts.
- **Medical Coverage Months:** This variable has the widest spread and the highest number of upper outliers. The highest variance **(94,925,221,933.13)** indicates substantial variation in coverage months.
- **Service Count:** Several upper outliers indicate that a few healthcare groups record exceptionally high service utilization. The variance **(5,679,367,784.22)** confirms considerable variability.
- **Average Services per Member:** Most values are concentrated within a smaller range with fewer extreme outliers. The lowest variance **(8.90)** indicates relatively consistent utilization per member.
- **Overall Observation:** All four box plots contain upper outliers, showing that a small number of healthcare groups have unusually high values compared to the majority of the dataset.

9.3 MEASURE OF DISPERSION: STANDARD DEVIATION

VARIABLES	STANDARD DEVIATION
member_count	12,127.34
medical_coverage_months	308,099.37
service_count	75,361.58
avg_services_per_member	2.98

INTERPRETATION

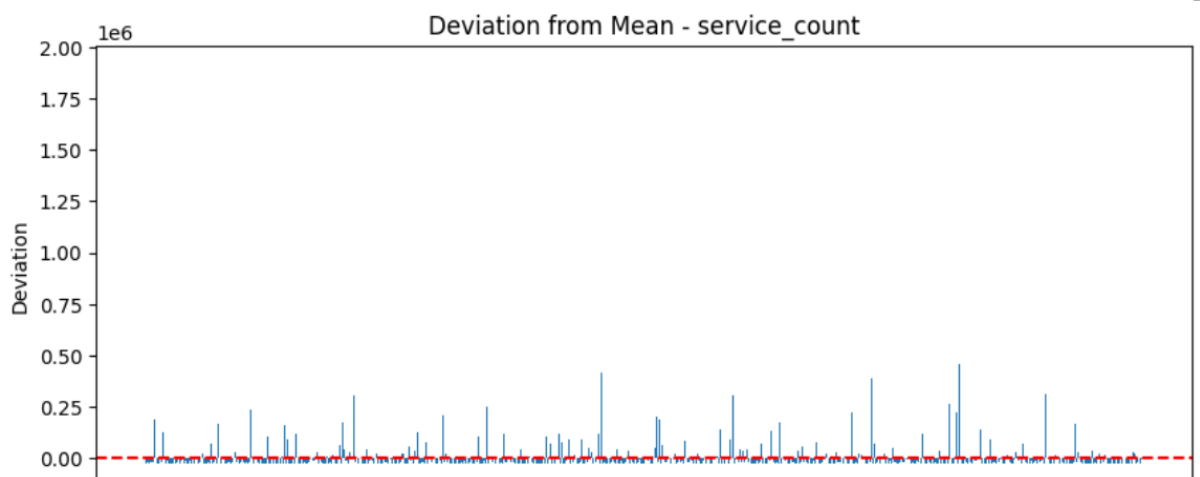
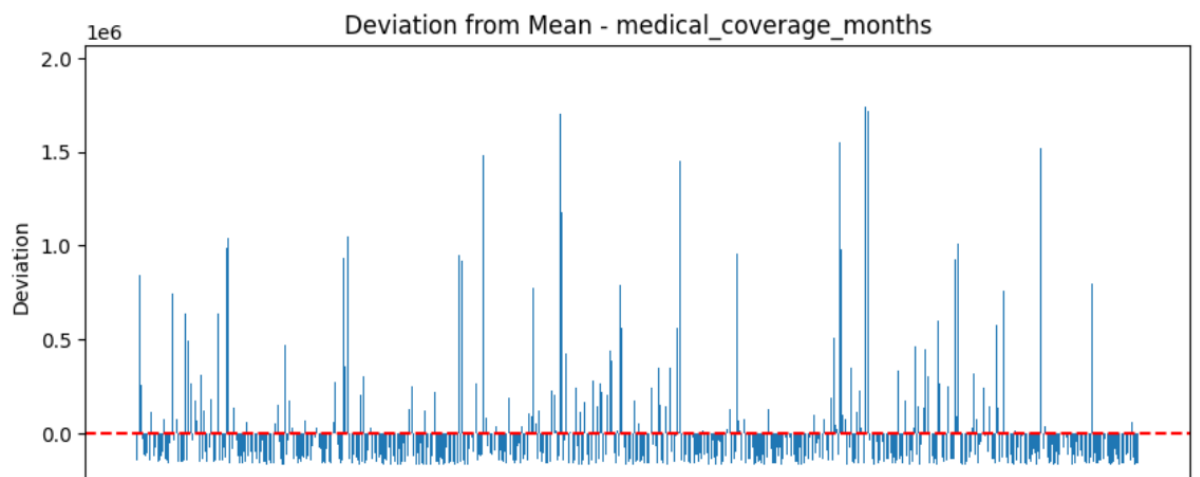
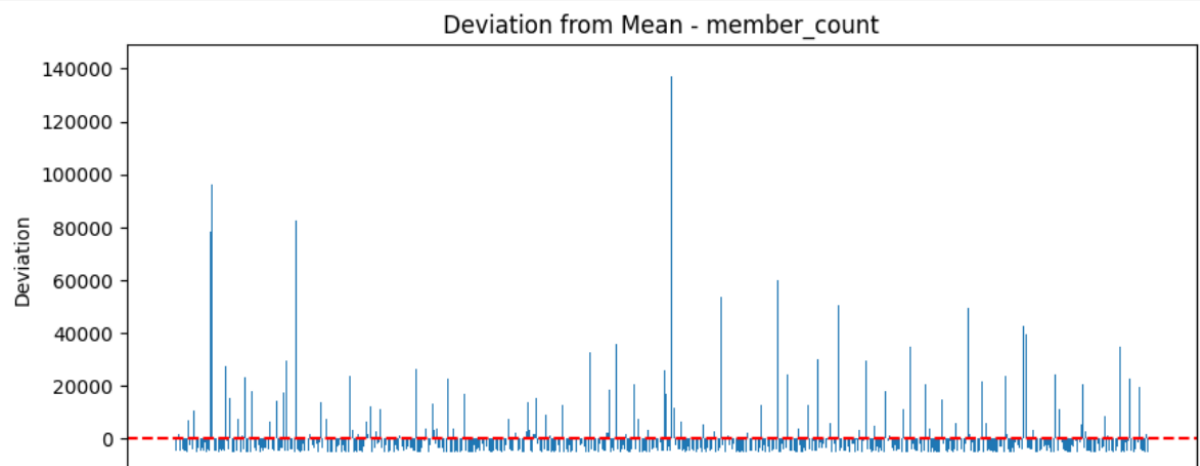
- The **Standard Deviation** measures how much the **data varies** from the **mean**.
- A **higher** standard deviation indicates greater **variability**, while a **lower** standard deviation indicates more **consistent** values.

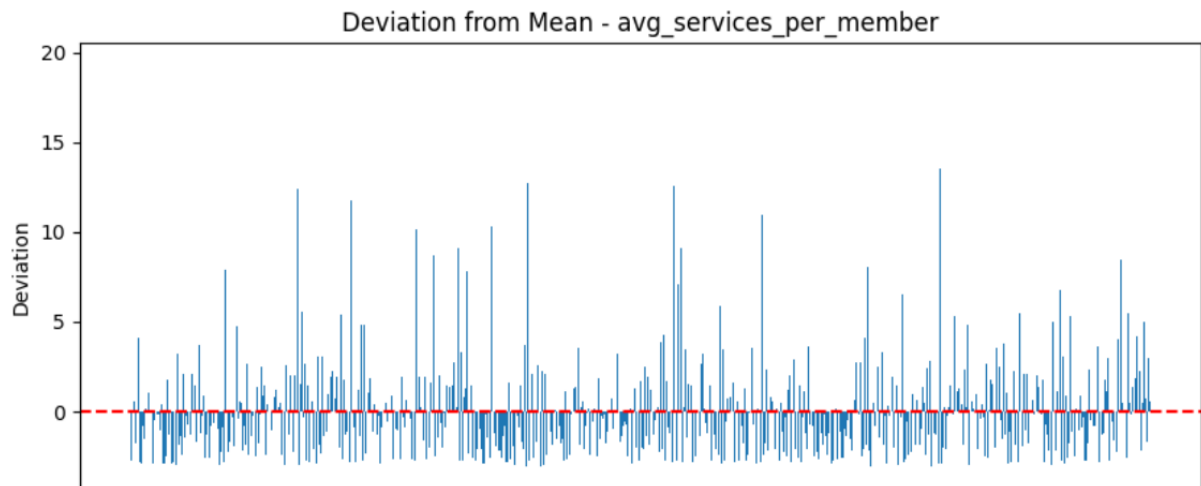
INSIGHTS

- **Member Count:** Shows moderate variation across healthcare groups with a standard deviation of **12,127.34**, indicating noticeable differences in member populations.
- **Medical Coverage Months:** Has the highest standard deviation **(308,099.37)**, indicating the greatest variability in healthcare coverage months.
- **Service Count:** Shows high variation in healthcare service utilization with a standard deviation of **75,361.58** across healthcare groups.
- **Average Services per Member:** Has the lowest standard deviation **(2.98)**, indicating relatively consistent service utilization per member.

- **Overall Observation:** Medical Coverage Months and Service Count show the greatest variability, while Average Services per Member is the most consistent variable.

9.4 DEVIATION FROM MEAN CHART





INTERPRETATION

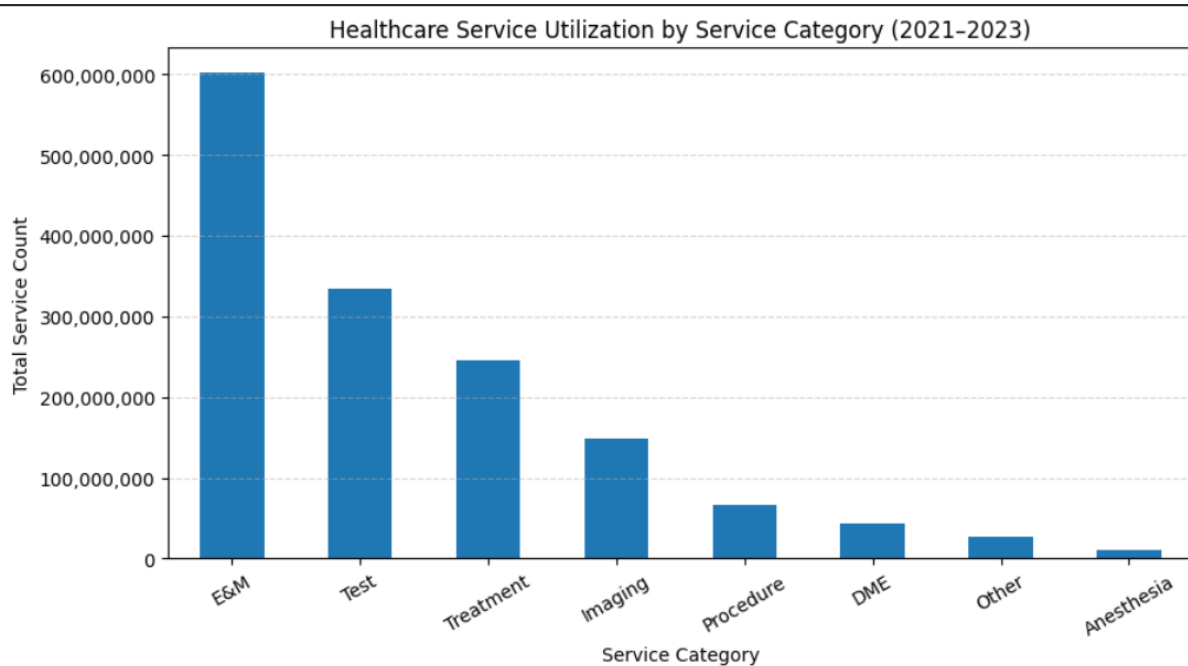
- The **Deviation from Mean** chart shows how far each observation is from the **average** value.
- Values **above** the **red line** are **higher** than the **mean**, while values **below** the **red line** are **lower** than the mean. Larger deviations indicate greater variation in the data.

INSIGHTS

- **Member Count:** Most observations are close to the **mean**, while a few healthcare groups have substantially **higher member counts**.
- **Medical Coverage Months:** Shows the **largest deviations** from the mean, indicating considerable variation in healthcare coverage across groups.
- **Service Count:** Most healthcare groups have **lower service utilization**, while a few groups record significantly higher service usage.
- **Average Services per Member:** Deviations are relatively **small**, indicating more consistent healthcare utilization **per member**.
- **Overall Observation:** All variables contain a few observations that deviate substantially from the mean, suggesting **uneven** healthcare utilization across **different groups**.

10. DATA VISUALISATION

10.1 To identify the most utilized healthcare service categories across California between 2021 and 2023

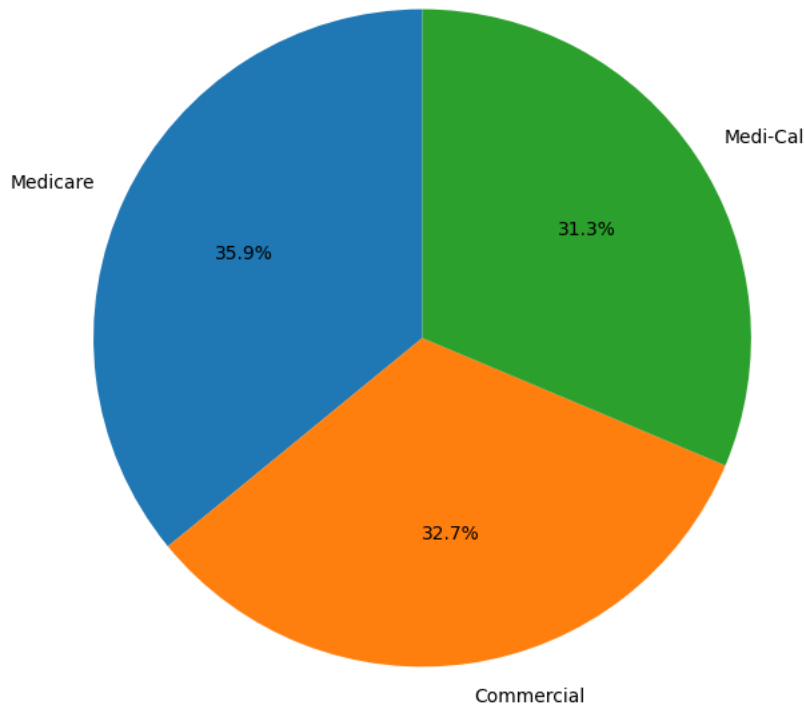


INTERPRETATION & INSIGHTS

- **Most Utilized Service:** This **Bar chart** shows the distribution of healthcare service utilization across different service categories and it shows the **E&M** records the **highest** healthcare service utilization between **2021** to **2023**, followed by **Test** and **Treatment**, indicating the **greatest demand** for these services.
- **Moderately Utilized Services:** **Imaging** shows **moderate** utilization, while **Procedure** and **DME** contribute a comparatively **smaller** share of total healthcare services.
- **Least Utilized Services:** **Other** and **Anesthesia** record the **lowest** service utilization, reflecting **limited** demand compared to the major service categories.
- **Overall Observation:** Healthcare service utilization is **unevenly** distributed, with **E&M** accounting for a significantly larger share than the remaining service categories.
- **Business Insight:** Identifying **high-demand** service categories helps healthcare organizations optimize resource allocation, workforce planning, and service delivery.

10.2 To compare healthcare service utilization across different insurance payer types to understand variations in service demand

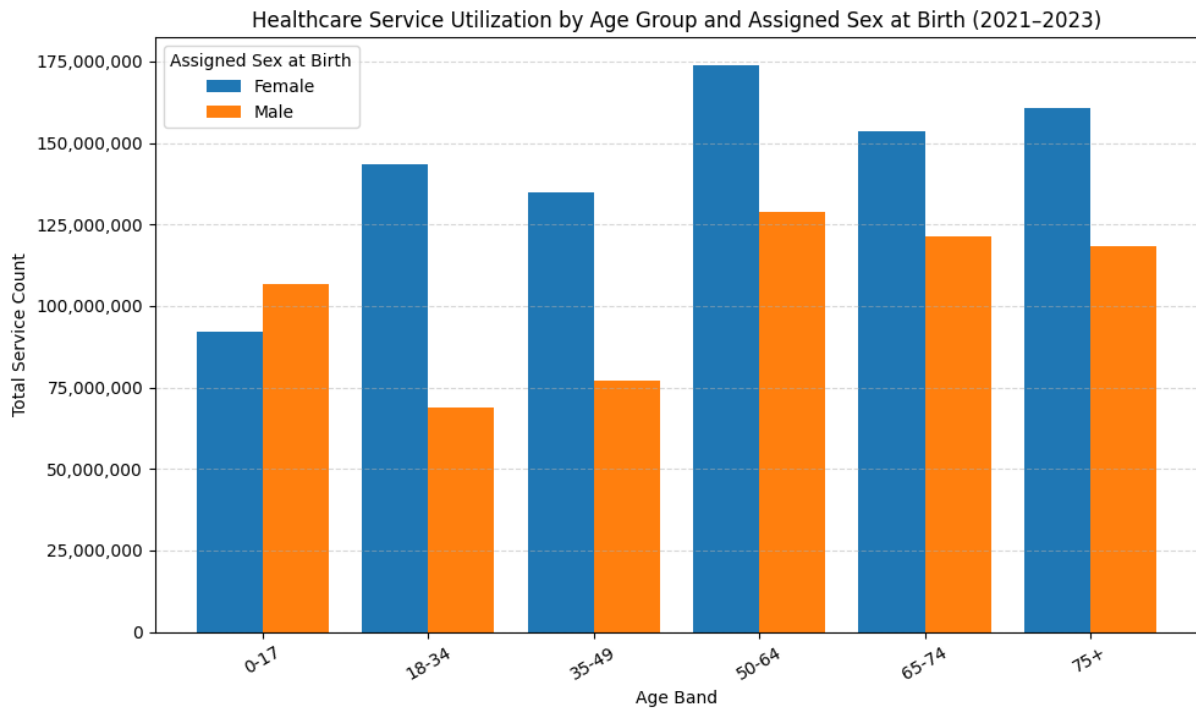
Healthcare Service Utilization by Payer Type (2021–2023)



INTERPRETATION & INSIGHTS

- **Healthcare Utilization Distribution:** The **pie chart** shows the **distribution** of total healthcare service utilization across the three major **payer types** between **2021** to **2023**.
- **Highest Utilization:** **Medicare** accounts for the **largest** share of healthcare service utilization at **35.9%**, indicating the **highest** service demand among all payer types.
- **Balanced Payer Contribution:** **Commercial** insurance contributes **32.7%**, followed by **Medi-Cal** at **31.3%**, showing that healthcare utilization is relatively **balanced** across the three payer groups.
- **Overall Observation:** The difference between the highest and lowest payer share is only **4.6 percentage** points, indicating that no single payer type overwhelmingly **dominates** healthcare service utilization.
- **Business Insight:** Understanding **payer-wise utilization** helps healthcare organizations optimize resource allocation, develop insurance-specific strategies, and support informed healthcare planning.

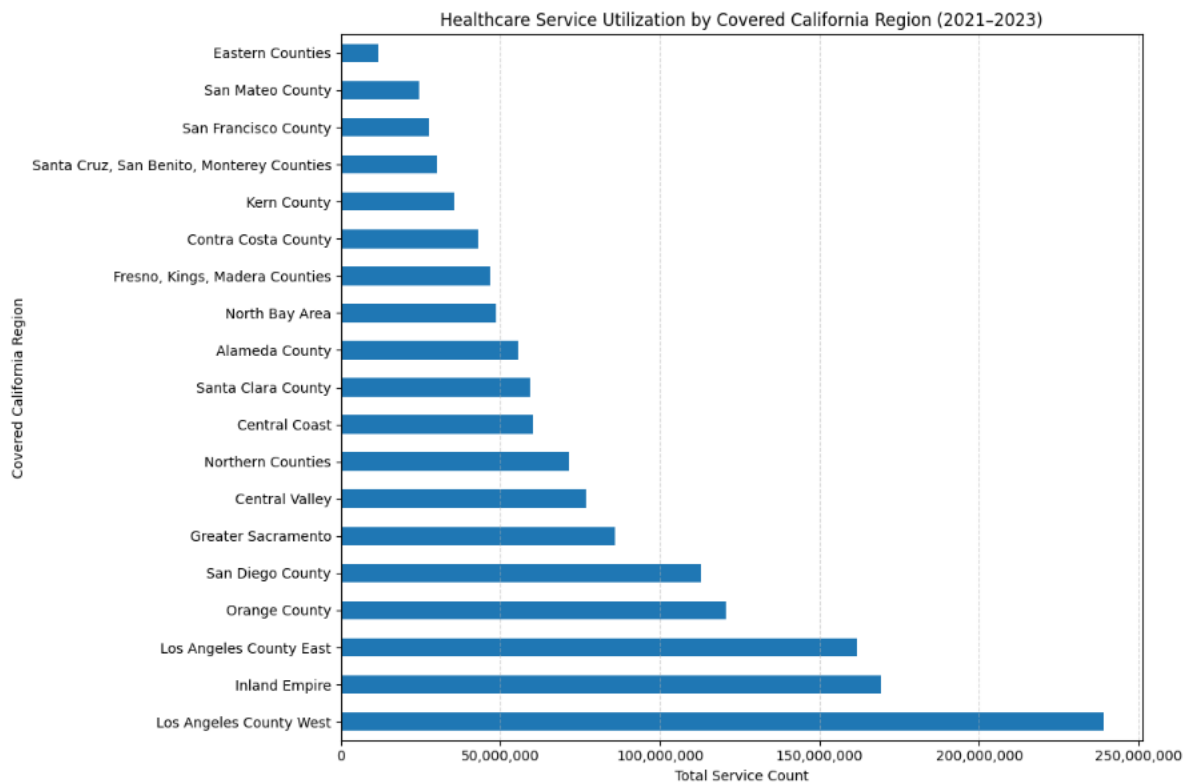
10.3 Demographic Analysis of Healthcare Service Utilization by Age Group and Assigned Sex at Birth



INTERPRETATION & INSIGHTS

- **Healthcare Utilization Across Age Groups:** This **clustered bar chart** compares **total healthcare service** utilization across **different age groups** between **2021** to **2023** for **females** and **males**.
- **Variation by Assigned Sex at Birth:** **Female** beneficiaries record **higher** healthcare service utilization in five of the six age groups, while **males** show slightly **higher** utilization only in the **0–17** age group.
- **Highest Healthcare Utilization:** The **50–64** age group records the **highest** healthcare service utilization for both **females** (approximately **167 million** services) and **males** (approximately **122 million** services), indicating the greatest demand among **middle-aged adults**.
- **Overall Observation:** Healthcare service utilization generally increases from **younger to middle-aged** groups and remains relatively high among **older adults**, with **female** utilization consistently exceeding **male** utilization in most age groups.
- **Business Insight:** Understanding utilization patterns by **age group** and **assigned sex at birth** helps healthcare providers optimize resource allocation, design targeted healthcare programs, and improve service planning for high-demand populations.

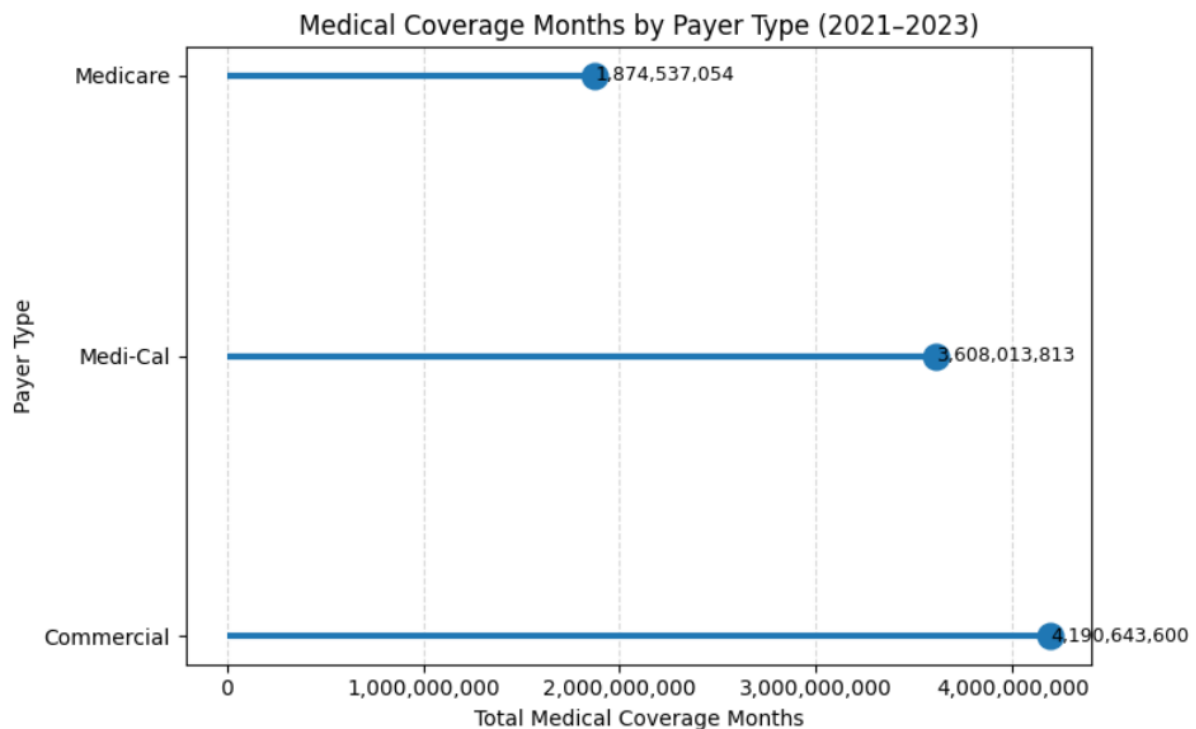
10.4 Regional Analysis of Healthcare Service Utilization by Covered California Region



INTERPRETATION & INSIGHTS

- **Regional Healthcare Utilization:** This **horizontal bar chart** compares total healthcare service utilization across **Covered California regions** between **2021 to 2023**, highlighting regional differences in healthcare demand.
- **Highest Healthcare Utilization:** **Los Angeles County West** records the **highest** healthcare service utilization at approximately **243 million** services, followed by **Inland Empire** (approximately **168 million**) and **Los Angeles County East** (approximately **160 million**).
- **Lowest Healthcare Utilization:** **Eastern Counties** record the **lowest** healthcare service utilization at approximately **12 million** services, followed by **San Mateo County** (approximately **25 million**) and **San Francisco County** (approximately **28 million**).
- **Overall Observation:** Healthcare service utilization varies considerably across regions, with **Los Angeles County West** recording nearly **twenty times** more services than **Eastern Counties**.
- **Business Insight:** Identifying regional utilization patterns helps **healthcare providers** and **policymakers** allocate resources **efficiently**, **strengthen** healthcare **planning**, and **prioritize** investments in high-demand regions while improving access in **lower-utilization areas**.

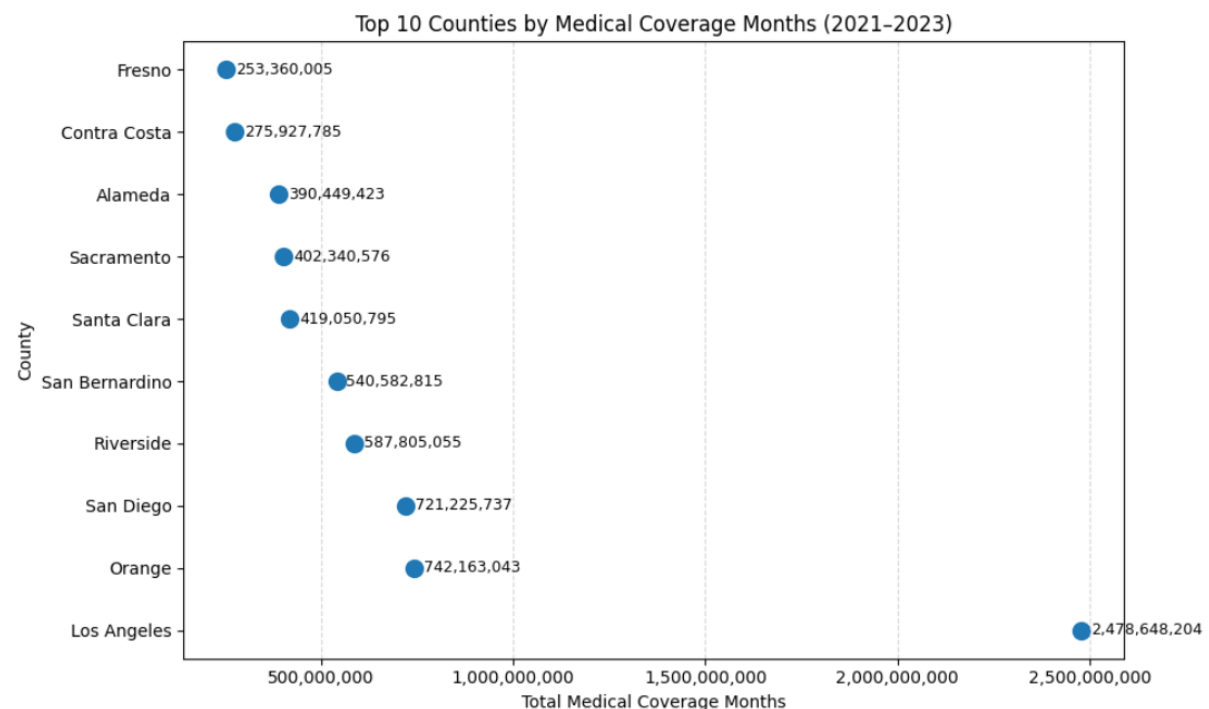
10.5 Insurance Coverage Analysis By Payer Type



INTERPRETATION & INSIGHTS

- **Insurance Coverage Pattern by Payer Type:** This **lollipop chart** compares the **total medical coverage months** across **different payer types** between **2021 to 2023**, highlighting differences in insurance coverage among **Commercial, Medi-Cal, and Medicare plans**.
- **Highest Medical Coverage:** **Commercial** insurance records the **highest** total medical coverage months (approximately **4.19 billion**), followed by **Medi-Cal** (approximately **3.61 billion**), indicating that these payer types account for the **largest share** of health insurance coverage during the study period.
- **Lowest Medical Coverage:** **Medicare** records the lowest total medical coverage months (approximately **1.87 billion**) among the **three payer types**, reflecting a comparatively **smaller** share of total insurance coverage.
- **Business Insight:** Analyzing medical coverage months across different payer types helps healthcare **administrators** and **policymakers** understand **insurance coverage** distribution, optimize healthcare resource planning, allocate resources effectively, and support informed policy decisions to improve healthcare accessibility.

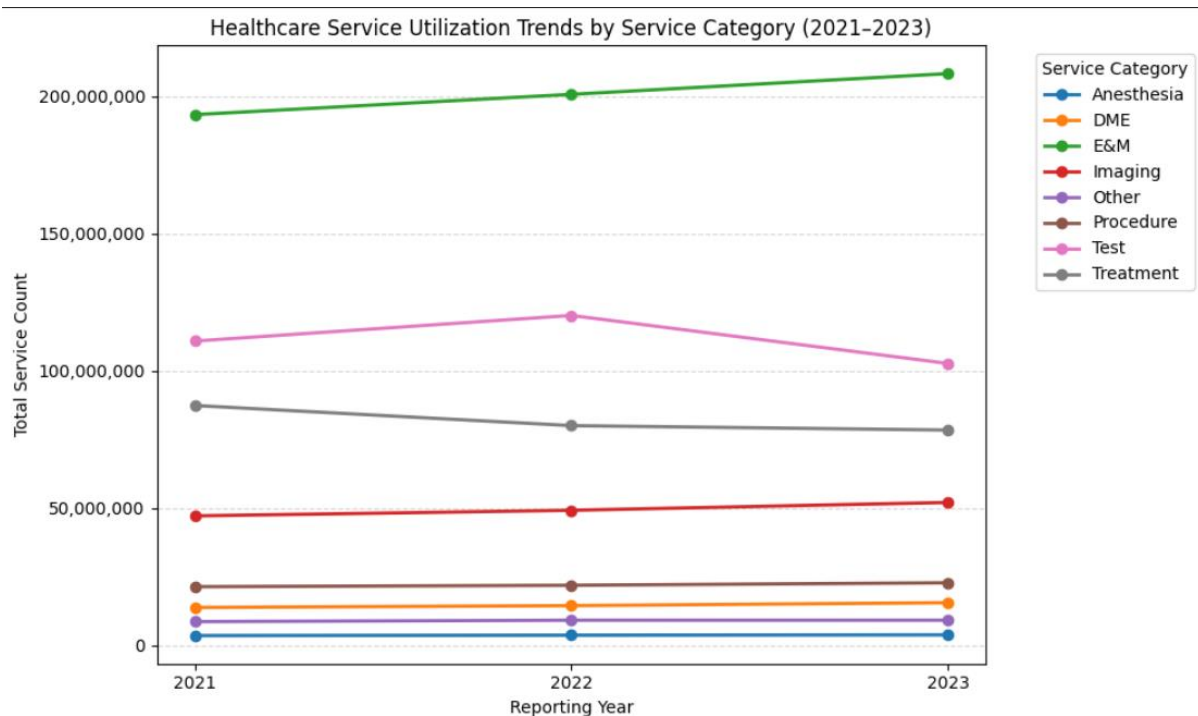
10.6 Insurance Coverage Analysis by County



INTERPRETATION & INSIGHTS

- **County-wise Insurance Coverage:** This **Cleveland dot plot** compares the **total medical coverage months** across the top **10 counties** in **California** between **2021 to 2023**, highlighting differences in insurance coverage among counties.
- **Highest Medical Coverage:** **Los Angeles County** records the **highest** total medical coverage months (approximately **2.48 billion**), followed by **Orange County** (approximately **742 million**) and **San Diego County** (approximately **721 million**), indicating a **larger** covered population and greater healthcare coverage in these counties.
- **Lower Medical Coverage:** Counties such as **Fresno** (approximately **253 million**), **Contra Costa** (approximately **276 million**), **Alameda** (approximately **390 million**), and **Sacramento** (approximately **402 million**) report comparatively **lower** medical coverage months among the top 10 counties, reflecting differences in population size and insurance coverage across counties.
- **Business Insight:** Identifying counties with **higher** and **lower** medical coverage months helps healthcare **administrators** and **policymakers** understand **regional** insurance coverage patterns, optimize healthcare resource allocation, and support data-driven planning for future healthcare programs.

10.7 To identify healthcare utilization trends over the period 2021–2023 and examine changes in service demand across different healthcare service categories



INTERPRETATION & INSIGHTS

- **Healthcare Utilization Trends:** The **multi-line chart** illustrates healthcare **service utilization trends** across **different service** categories between 2021 to 2023, highlighting how healthcare service demand changed over the three-year period.
- **Increasing Service Demand:** E&M records the **highest** healthcare service utilization throughout the study period and shows a steady increase from 2021 to 2023, indicating **consistent** growth in demand for evaluation and **management services**.
- **Changing Service Utilization:** Test services increase from 2021 to 2022 before declining in 2023, while Treatment services show a **gradual** decline over the **three-**year period. Imaging shows a **steady** increase, whereas Procedure, DME, Other, and Anesthesia remain relatively **stable** with only **minor** changes.
- **Business Insight:** Analyzing healthcare utilization **trends** across different service categories helps healthcare **administrators** and **policymakers** identify changes in service demand, optimize healthcare resource allocation, improve operational planning, and support data-driven decision-making for future healthcare service delivery.

11. BUSINESS STRATEGY RECOMMENDATIONS

- **Prioritize** resources for **high-demand** healthcare services, particularly **Evaluation & Management (E&M)** and **Test services**.
- Develop **payer-specific strategies** to improve healthcare service planning across Medicare, Commercial, and Medi-Cal.

- **Focus** healthcare programs on **high-utilization age groups and regions** to improve service accessibility.
- **Strengthen** resource allocation in **high-demand regions** while improving healthcare access in **lower-utilization** areas.
- **Monitor** healthcare utilization **trends regularly** to support data-driven planning and efficient **decision-making**.

12. CONCLUSION

This project analyzed **healthcare service utilization** data from **California** between **2021** to **2023** using **Python**. Through data **preprocessing**, **feature engineering**, **statistical analysis**, and **visualization**, the study identified utilization patterns across **service categories**, **payer types**, **demographic groups**, **regions**, **insurance coverage**, and **yearly trends**. The findings provide **valuable** insights that can **support** effective **resource allocation**, **healthcare planning**, and informed decision-making, demonstrating the **importance** of **data analytics** in improving healthcare service delivery.